Voice mismatch in antecedent-contained deletion

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November 6, 2026

1 Ellipsis

Under what conditions can certain phrases be left unspoken? When are we allowed to elide VPs?

(1) Jorge can eat baklava. Rafa can't <eat baklava>.

Texts enclosed in angle brackets <> are the elided portion of the sentence.

Focusing on VP-ellipsis, we could characterize the conditions as *identity* with an antecedent VP.

But what happens when a trace or pronoun is in the ellipsis site?

(2) Jorge can eat his$_j$ dinner. Rafa can't <eat his$_r$ dinner>.

What makes the pronoun *his*$_j$ identical to *his*$_r$ when they refer to different individuals? Challenges for formulating what exactly *identity* between two VPs means come from mismatches between the antecedent and elided VPs.

Another apparent mismatch is known as Antecedent Contained Deletion (ACD), where the elided VP is seemingly contained within the VP that should serve as its antecedent (Bouton 1970).

(3) I met every linguist that you did <meet>.

The elided VP cannot be identical to any apparent antecedent VP, otherwise the VP that was elided would need to be *meet every linguist that you did meet every linguist that you did meet...* How can we get identity between the two VPs?

Yet another apparent mismatch between two VPs is alternating **voice**. VP-ellipsis is still licensed when one verb is in active voice and the other is in passive voice (Merchant 2013).

(4) a. The janitor must remove the trash whenever it is apparent that it should be <removed>.

b. The system can be used by anyone who wants to <use it>.

Examples are from Merchant (2013). (4a) shows an example with an active voice antecedent and a passive voice ellipsis site. (4b) shows an example with a passive voice antecedent and an active voice ellipsis site.

What makes the active voice verb identical to the passive voice verb when active and passive have different argument structures?

Allow me to introduce another observation: Voice mismatches are not allowed under ACD.

(5) a. * The janitor must remove the trash that should be <removed>.

b. * The janitor must remove every piece of trash that should be <removed>.

c. * The professor must assign the problems that should be <assigned>.

d. * The professor must assign every problem that should be <assigned>.

ACD is VP ellipsis. What is it about ACD that does not permit identity between voice mismatches? The ungrammaticality of voice-mismatch ACD comes from additional requirements that ACD configurations place on identity.

Different formulations of the identity condition for ellipsis can account for the trace/pronouns problem, the ACD problem, and the voice-mismatch problem. The goal here is to formulate a theory of ellipsis that accounts for the following:

1. Ellipsis is licensed in ACD
2. Ellipsis is licensed in normal voice mismatches
3. Ellipsis is not licensed in ACD voice mismatches

Preview: I end up with a picture of ellipsis where ACD is licensed via the identity condition. Non-identity in voice mismatches is able to be rescued via mechanisms that change how traces are interpreted. These same mechanisms are not available for ACD, therefore ACD voice mismatches are ungrammatical, with an exception.

2 A semantic identity condition

We begin with a preliminary identity condition for ellipsis following Heim (1996).

(6) A VP can be elided iff it is semantically identical to an antecedent VP.

a. VP_1 is semantically identical to VP_2 if for every assignment function g, $[\![VP_1]\!]^g = [\![VP_2]\!]^g$

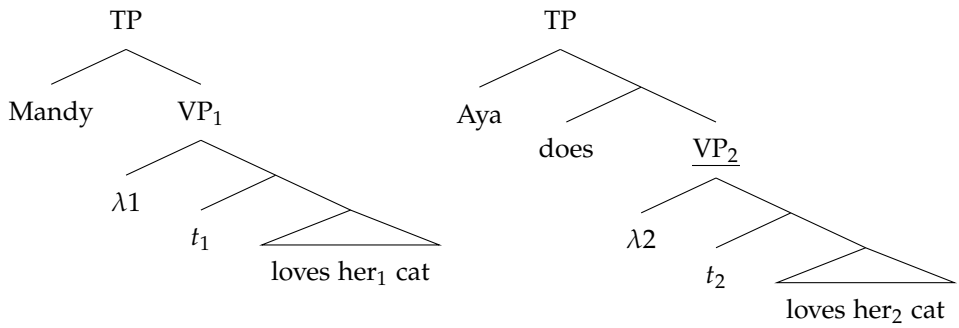
We also need to assume a condition of *No Meaningless Co-indexation* so that variables do not share the same index unless they are bound by the same binder (Heim 1996).

With the above condition, under a standard theory of traces and pronouns as indexed variables, traces/pronouns in an ellipsis site are identical only if they bear the same index or are bound inside of the VP.

(7) Mandy^1 loves her$_1$ cat. Aya^2 does <love her$_2$ cat> too.

Even though the pronoun in the elided VP does not have the same index as the pronoun in the antecedent VP, we do get identity with an LF that includes the binder.

(8) a.



b. Identity:
For every assignment function g,
$[\![\lambda 1\ t_1$ loves her$_1$ cat$]\!]^g = [\![\lambda 2\ t_2$ loves her$_2$ cat$]\!]^g$

However, sometimes a binder outside of the VP can bind a trace/pronoun within the ellipsis site.

(9) Seva[1] wanted James to write his$_1$ paper. Eyal[2] wanted Karo to <write his$_2$ paper>.

The pronouns fail identity and the binders are outside of the VPs, therefore the identity condition in (6) does not predict that ellipsis is licensed here.

So the identity condition in (6) must be altered to allow us to search an area larger than the ellipsis site. Here I follow Rooth (1992), Heim (1996), and Takahashi & Fox (2005) and rewrite the identity condition based on a *parallelism* condition.

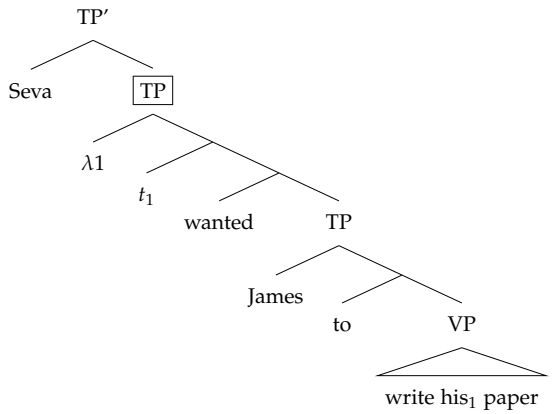
(10) A phrase E can be elided iff there is a constituent PD that reflexively dominates E and satisfies Parallelism.

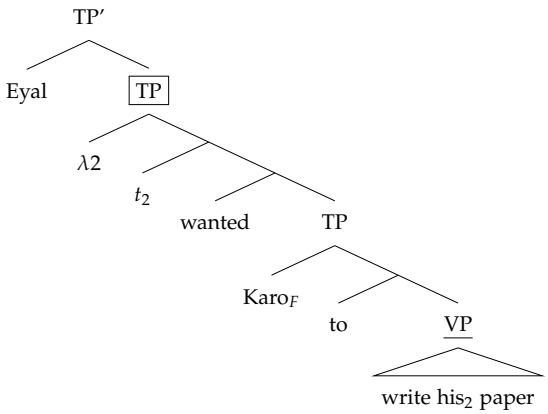
Parallelism is defined as follows:

(11) Parallelism
A constituent PD (Parallelism Domain) satisfies Parallelism if PD is semantically identical to an antecedent phrase A modulo focus marked constituents.
   a. PD is semantically identical to A modulo focus if for every assignment function g, $[\![\text{A}]\!]^g \in [\![\text{PD}]\!]^g_F$

The PD can be the elided VP itself, or a larger constituent. If the ellipsis site contains a trace, then the PD must be large enough to contain the binder as well (Takahashi & Fox 2005, Hartman 2011).

(12) a.





b. Parallelism:
$[\![\lambda 1\ t_1$ wanted James to write his$_1$ paper$]\!]^g \in$
$[\![\lambda 2\ t_2$ wanted Karo$_F$ to write his$_2$ paper$]\!]^g_F$

The boxed nodes are the PD and its antecedent that are under consideration by the identity condition. The underlined node is the elided VP.

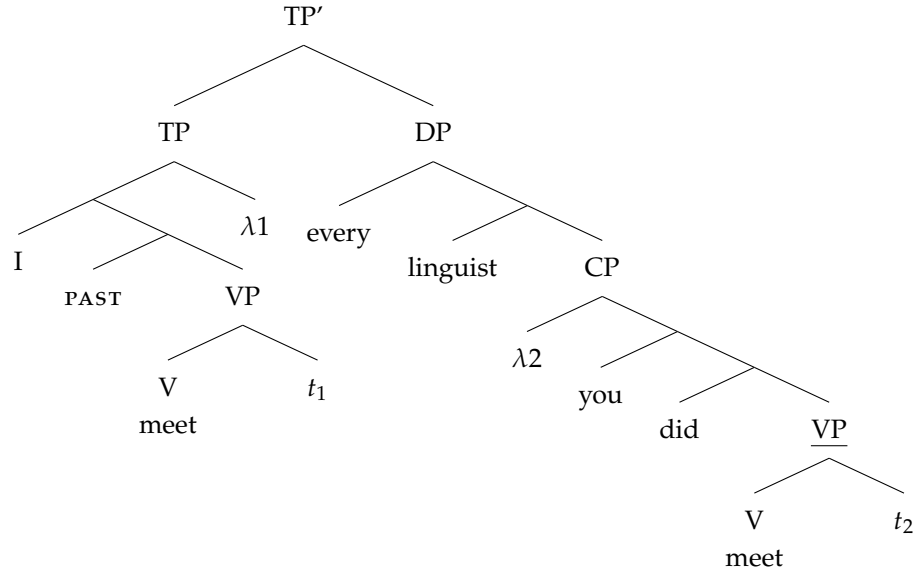
Focus on constituents is marked with a subscript *F*.

3 Antecedent Contained Deletion

A common view for resolution of ACD proposes that a syntactic operation, Quantifier Raising (QR), can generate a structure that moves the object quantifier, which contains the elided VP, to a position outside of the antecedent VP (Sag 1976, Larson & May 1990).

(13) I met every linguist you did.

a.

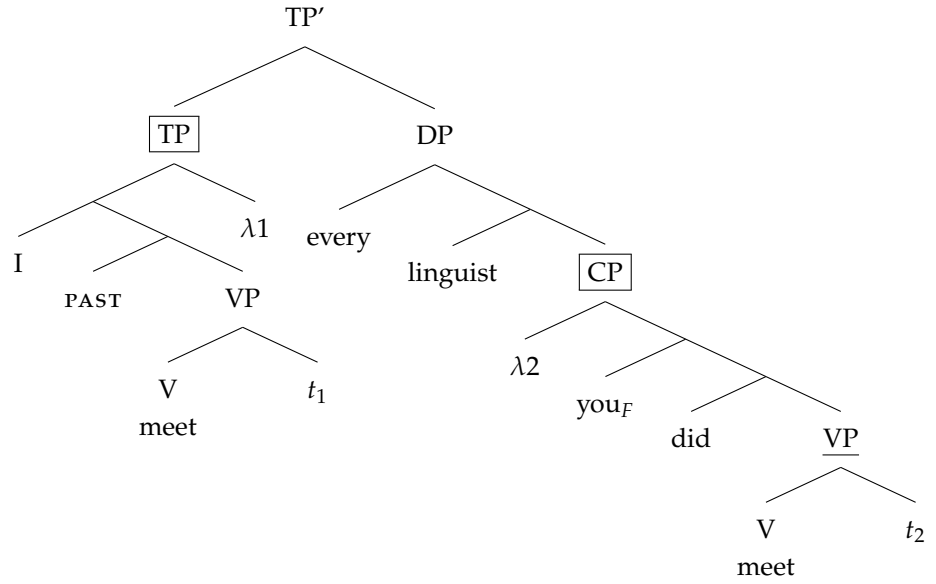


Since the elided VP and the antecedent VP contain traces with different indices, we are forced to look at a larger constituent to find a valid PD.

With focus marking on the correct constituents, we find a PD that satisfies the identity condition in (10). This is shown in (14).

(14) I met every linguist you$_F$ did.

a.



b. Parallelism: $[\![\lambda 1$ I PAST meet $t_1]\!]^g \in [\![\lambda 2$ you$_F$ did meet $t_2]\!]^g_F$

Now we have an identity condition that works for ACD. Next we need to allow for identity to hold for normal voice mismatch ellipsis.

4 Voice mismatch ellipsis

The problem of voice mismatch VP ellipsis actually derives from the observation that VP ellipsis permits voice mismatches while larger ellipses, such as sluicing, do not permit voice mismatches (Merchant 2013).

(4) a. The janitor must remove the trash whenever it is apparent that it should be <removed>. (Merchant 2013)

b. The system can be used by anyone who wants to <use it>. (Merchant 2013)

In contrast, voice mismatches are unacceptable in other types of ellipsis, such as sluicing.

(15) a. * Joe was murdered, but we don't know who. (Merchant 2013)

b. * Someone murdered Joe, but we don't know who by. (Merchant 2013)

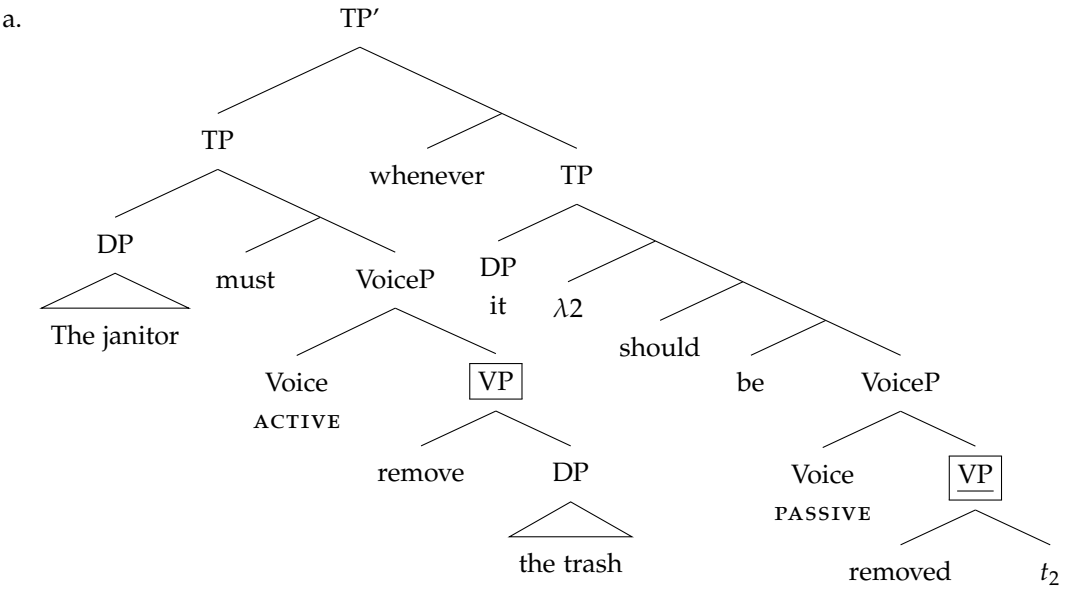
Merchant (2013) argues that the availability of voice mismatches in VP ellipsis can be explained by a size difference between the ellipsis site in VP ellipsis versus other types of ellipsis such as sluicing. He

proposes that VoiceP is outside of the ellipsis site in VP ellipsis but inside of the ellipsis site in sluicing. I'll adopt this view of VoiceP and size of ellipsis.

A structure for (4a) is shown in (16), with the possible PD.

(16) The janitor must remove the trash whenever it should be.

a.



In order to maintain the idea from Merchant (2013) about exclusion of VoiceP in VP ellipsis, the PD must be lower than VoiceP. This is because the Voice head is either impossible to focus or, even if ACTIVE/PASSIVE is focused, there will be no antecedent node that is semantically identical.

For instance, under a view that the voice head introduces or closes arguments, then the lexical entries for the voice heads in an event semantics should look like the following:

See Kratzer (1996) on this view of Voice.

(17) $[\![\text{ACTIVE}]\!]^g = \lambda P_{\langle v,t\rangle}.\lambda e_v.\lambda x_e.Agent(x)(e) \wedge P(e)$

(18) $[\![\text{PASSIVE}]\!]^g = \lambda P_{\langle v,t\rangle}.\lambda e_v.\exists x[Agent(x)(e) \wedge P(e)]$

Where active introduces the external argument while passive existentially closes the external argument. Any position where ACTIVE is replaced with PASSIVE or vice-versa will result in an uninterpretable structure where there is either too little or too many open argument slots.

So the possible PD in the voice mismatch sentences is limited to just the VP. But there is no semantic identity, since in (16) the elided VP contains a trace and the antecedent VP contains a DP *the trash*.

There are two possible mechanisms that I can imagine to rescue non-identity here.

- Merchant (2013) suggests that the content of the trace left by *it* must be understood in the ellipsis site.

My interpretation of this is that the semantic value of a trace that is left by A-movement of a referential term, such as *it*, can be interpreted as the referent itself. If the semantic value of the trace is also *the trash*, then identity is satisfied.

This is a stipulation, however. Some support comes from evidence that traces of A-movement do not count for Parallelism by Messick & Thoms (2016), investigating a different problem.

- Focus on the head of a movement chain may percolate down to traces in the movement chain.

This is due to examples with contrastive focus that permit the voice mismatch ellipsis.

(19) The janitor must remove the trash whenever the recycling$_F$ must be.

Here we can say that if the head of a movement chain is focused, then the trace can be optionally focused as well, following Sauerland (1998), Takahashi & Fox (2005), and others. This means that since *the recycling* is focused, then the trace left by the passive movement can also be focused. Then we end up with a PD that is semantically identical modulo focus to the antecedent VP.

(20) $[\![\text{remove the trash}]\!]^g \in [\![\text{remove } t_{2F}]\!]^g_F$

These are two possible rescue mechanisms for identity in voice mismatch ellipsis. Next, let's examine what the identity condition does when applied to voice mismatch ACD.

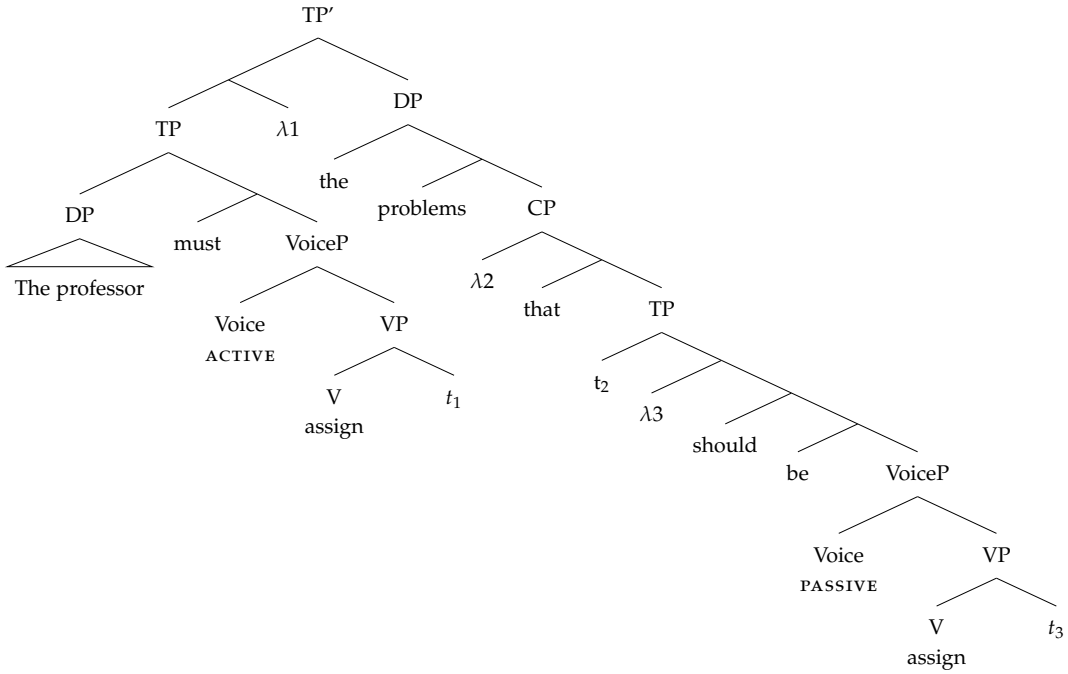
Another way of thinking about this suggestion might be syntactic reconstruction. But this would force the scope of the DP to be very low. This doesn't seem to be the case if we put a quantificational DP. In the example below, wide-scope of the DP over the modal is possible.
(i.) The janitor must remove the trash whenever 3 recycling bins must be.

5 Voice mismatch ACD

Finally, we arrive at the main puzzle. Let's consider a possible structure for the sentence in (21). Note that the good version of this sentence is *The professor must assign the problems that should be assigned*.

(21) * The professor must assign the problems that should be

a.



The movements in the structure are first relative pronoun from the object to SpecTP to satisfy EPP, where it leaves a trace and inserts a binder below it. Then the relative pronoun movement to SpecCP, where it gets converted to a binder that binds the trace of its last movement by Predicate Abstraction.

For identity to hold between the two VPs, a PD must be found that satisfies Parallelism. However, since any PD larger than VP will differ in voice from any antecedent, the PD must only be VP.

Both potential strategies for rescuing identity between the VPs fail here.

- Interpreting the trace t_3 with the content of the moved expression.

This will not work because the trace was left by movement of a relative operator. The relative operator is not referential and its semantic contribution is to bind the trace from the edge of the relative clause.

- Focusing the head of the relative clause.

The strategy of focusing the head of the movement chain will also not work for ACD. This is because the trace can only be focused if

the focused moved item is related directly to the trace, such as with the movement involved in passivization. The difference is that this relative clause is externally headed. *The problems* never originates in the relative clause. This is supported by arguments from Hulsey & Sauerland (2006) that the relative clause structure required for ACD is a matching structure, not a raising structure.

The argument for a matching structure for ACD relates to Fox (2002) where under a copy theory of movement ACD resolution is QR with extraposition such that the relative clause is Late Merged into the higher copy. Extraposition forces a matching structure, and thus ACD does as well. However, Hulsey & Sauerland (2006) point out there are problems for this assumption related to binding.

Since the ellipsis site contains a trace that cannot be focused, we are forced to select a PD that contains the binder of the variable. But any PD larger than VP will also include VoiceP and thus will not be identical with any antecedent phrase. Thus the identity condition in (10) fails and cannot be rescued in the same way as normal voice mismatch ellipsis.

6 Indirect Parallelism

However, there are some ACD sentences where voice-mismatches do seem to be permitted.

(22) a. The professor should have assigned a problem that wasn't <assigned>.

b. The professor should have assigned the problem that wasn't <assigned>.

The configurations where ACD voice-mismatches are allowed all seem to include elements of negation (*wasn't*) and counterfactual modals (*should have*). The mechanism that rescues identity in these cases is Indirect Parallelism through accommodation of another proposition (Fox 1999), defined below.

(23) Indirect Parallelism
A constituent *PD* satisfies Indirect Parallelism if *PD* is semantically identical modulo focus to an accommodated antecedent *A'*.

a. Where an accommodated antecedent *A'* is a proposition that is entailed by an antecedent *A* together with the common ground.

b. Accommodation is triggered when the *PD* contains accommodation-seeking-material (pronounced constituents that are non-F-marked in *A*).

The antecedent *A* is *The professor should have assigned a problem*. The counterfactual inference from this is that *The professor didn't assign a problem*, which along with a common ground assumption that the professor is the one who assigns problems, entails the passive proposition:

(24) $\exists y[problem(y) \wedge \neg\exists e\exists x[Agent(x)(e) \wedge assign(e)(y)]$

Taking this as the accommodated antecedent A', ellipsis is licensed because a PD, which now includes VoiceP, is semantically identical modulo focus to A'.

Stockwell (2024) uses a similar mechanism for deriving contrasts between simpler voice mismatch ellipses, such as the following.

(25) a. This information should have been released, but Gorbachev didn't <release it>.

b. * This information was released, but Gorbachev didn't <release it>.

In fact, many of the examples of voice-mismatch ellipses from Merchant (2013) include counterfactual phrases such as *should have*, *could have*, *was to have been*... and negation. It may be that voice-mismatches are often cases where a mechanism like Indirect Parallelism supports licensing of ellipsis, but ACD configurations place additional stress on the availability of an accommodated antecedent.

I have assumed that voice-mismatches are fundamentally grammatical, i.e. that the identity condition holds and ellipsis is licensed. However, other views such as Arregui et al. (2006) suggest that voice-mismatches are ungrammatical. Arregui et al. (2006) propose the Recycling Hypothesis, appealing to processing and pragmatics to explain how a mismatched antecedent can be repaired. Formalizing where Indirect Parallelism can apply may relate to their discussion.

7 Conclusion

I presented an observation that voice mismatch ACD is an unacceptable type of ellipsis. This presented a challenge to formulate the proper identity condition for ellipsis. I showed the following:

- A semantic identity condition for ellipsis with Parallelism Domains is able to predict the licensing of ACD.
- With some additional mechanisms for traces, the identity condition also predicts the licensing of voice mismatch VP ellipsis.
- With some other assumptions, the identity condition is able to predict that voice mismatch ACD is not licensed.
- Voice mismatch ACD can be rescued when an accommodated antecedent is available.

Thanks to Elise Newman, David Pesetsky, and Danny Fox for discussions which this project grew from.

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